Project Exercise 1 statistics 2

📊 Dataset Description: BRFSS Subset (New York State) The resulting DataFrame, random_5k_rows, is a cleaned subset derived from the original 2021 Behavioral Risk Factor Surveillance System (BRFSS) survey data, conducted by the CDC. This specific dataset contains records exclusively for respondents from a single state, identified by code 36.0 (New York State).

The primary cleaning objective was to ensure data quality by removing rows with unknown, refused, or otherwise incomplete responses (coded as 7, 9, 77, 88, 99, or 'BLANK' based on the variables). After this extensive filtering, the dataset retains 27,166 valid records, after that we took randomly 5k rows.

_AGE80	גיל	רציף
SEXVAR	מין	קטגוריאלי
INCOME3	הכנסה שנתית	קטגוריאלי עם חשיבות לסדר
GENHLTH	בריאות כללית (תיאור עצמי)	קטגוריאלי
MENTHLTH	ימים בהם הורגש קושי נפשי	רציף
_RFSMOK3	מעשן/לא מעשן	קטגוריאלי
_BMI5	BMI	רציף
CVDINFR4	האם היה לך התקף לב ?	בינארי
ADDEPEV3	קיימת היסטוריה של דיכאונות	בינארי
TOLDHI3	היסטוריה של כולסטרול גבוה	בינארי

In [30]:
```
# --- Install required packages ---
!pip install pandas pyreadstat

# --- Import libraries ---
import pandas as pd
import pyreadstat

# --- Download BRFSS 2021 dataset from CDC and extract subset---
url = "https://www.cdc.gov/brfss/annual_data/2021/files/LLCP2021XPT.zip"

import requests, zipfile, io
r = requests.get(url)
z = zipfile.ZipFile(io.BytesIO(r.content))
z.extractall("brfss2021")

df, meta = pyreadstat.read_xport("brfss2021/LLCP2021.XPT ")

cols = [
    # Demographics
    "_AGE80", "SEXVAR", "INCOME3","_STATE",
    # Health behaviors & status
    "GENHLTH","MENTHLTH",
    "_RFSMOK3",
```

```
    # Anthropometrics
    "_BMI5",
    # Chronic conditions
    "CVDINFR4","DIABETE4",
    "ADDEPEV3", "TOLDHI3"
]
brfss_sub = df[cols].copy()
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-package
s (2.2.2)
Requirement already satisfied: pyreadstat in /usr/local/lib/python3.12/dist-pac
kages (1.3.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-
packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python
3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-p
ackages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dis
t-packages (from pandas) (2025.2)
Requirement already satisfied: narwhals>=2.0 in /usr/local/lib/python3.12/dist-
packages (from pyreadstat) (2.11.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packa
ges (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

In [31]:

```
brfss_sub = brfss_sub[
    (brfss_sub['INCOME3'] != 77 ) &
    (brfss_sub['INCOME3'] != 99) &
    (brfss_sub['INCOME3'] != 'BLANK')
]


brfss_sub1 = brfss_sub[brfss_sub['_STATE'] == 36]


brfss_sub1
```

Out[31]:

	_AGE80	SEXVAR	INCOME3	_STATE	GENHLTH	MENTHLTH	_RFSMOK3
254073	75.0	2.0	4.0	36.0	2.0	30.0	1.0
254074	18.0	2.0	9.0	36.0	2.0	1.0	1.0
254075	75.0	1.0	6.0	36.0	2.0	88.0	1.0
254077	56.0	1.0	5.0	36.0	4.0	3.0	2.0
254078	80.0	2.0	9.0	36.0	3.0	88.0	1.0
...	...	...	...	...	...	...	...
293163	33.0	2.0	4.0	36.0	3.0	15.0	2.0
293164	24.0	1.0	6.0	36.0	2.0	2.0	1.0
293165	40.0	2.0	7.0	36.0	2.0	17.0	1.0
293166	62.0	2.0	6.0	36.0	3.0	88.0	2.0
293167	36.0	2.0	NaN	36.0	1.0	88.0	9.0

30435 rows × 12 columns

In [32]:

```
brfss_sub1 = brfss_sub1.drop(columns = ['_STATE'])
brfss_sub1
```

Out[32]:

	_AGE80	SEXVAR	INCOME3	GENHLTH	MENTHLTH	_RFSMOK3	_BMI5
254073	75.0	2.0	4.0	2.0	30.0	1.0	2622.0
254074	18.0	2.0	9.0	2.0	1.0	1.0	2080.0
254075	75.0	1.0	6.0	2.0	88.0	1.0	2673.0
254077	56.0	1.0	5.0	4.0	3.0	2.0	3261.0
254078	80.0	2.0	9.0	3.0	88.0	1.0	2271.0
...	...	...	...	...	...	...	...
293163	33.0	2.0	4.0	3.0	15.0	2.0	4291.0
293164	24.0	1.0	6.0	2.0	2.0	1.0	2411.0
293165	40.0	2.0	7.0	2.0	17.0	1.0	2349.0
293166	62.0	2.0	6.0	3.0	88.0	2.0	2976.0
293167	36.0	2.0	NaN	1.0	88.0	9.0	NaN

30435 rows × 11 columns

In [33]:

```
brfss_sub2 = brfss_sub1[
    (brfss_sub1['GENHLTH'] != 7) &
    (brfss_sub1['GENHLTH'] != 9) &
```

```
    (brfss_sub1['GENHLTH'] != 'BLANK')
]


brfss_sub2
```

Out[33]:

	_AGE80	SEXVAR	INCOME3	GENHLTH	MENTHLTH	_RFSMOK3	_BMI5
254073	75.0	2.0	4.0	2.0	30.0	1.0	2622.0
254074	18.0	2.0	9.0	2.0	1.0	1.0	2080.0
254075	75.0	1.0	6.0	2.0	88.0	1.0	2673.0
254077	56.0	1.0	5.0	4.0	3.0	2.0	3261.0
254078	80.0	2.0	9.0	3.0	88.0	1.0	2271.0
...	...	...	...	...	...	...	...
293163	33.0	2.0	4.0	3.0	15.0	2.0	4291.0
293164	24.0	1.0	6.0	2.0	2.0	1.0	2411.0
293165	40.0	2.0	7.0	2.0	17.0	1.0	2349.0
293166	62.0	2.0	6.0	3.0	88.0	2.0	2976.0
293167	36.0	2.0	NaN	1.0	88.0	9.0	NaN

30378 rows × 11 columns

In [34]:

```
brfss_sub3 = brfss_sub2[
    (brfss_sub2['MENTHLTH'] != 77) &
    (brfss_sub2['MENTHLTH'] != 99) &
    (brfss_sub2['MENTHLTH'] != 88) &
    (brfss_sub2['MENTHLTH'] != 'BLANK')
]
brfss_sub3
```

Out[34]:

	_AGE80	SEXVAR	INCOME3	GENHLTH	MENTHLTH	_RFSMOK3	_BMI5
254073	75.0	2.0	4.0	2.0	30.0	1.0	2622.0
254074	18.0	2.0	9.0	2.0	1.0	1.0	2080.0
254077	56.0	1.0	5.0	4.0	3.0	2.0	3261.0
254082	72.0	1.0	7.0	4.0	30.0	1.0	NaN
254083	53.0	1.0	8.0	4.0	10.0	1.0	3839.0
...	...	...	...	...	...	...	...
293160	64.0	2.0	3.0	4.0	2.0	1.0	3411.0
293161	27.0	1.0	4.0	2.0	10.0	1.0	4161.0
293163	33.0	2.0	4.0	3.0	15.0	2.0	4291.0
293164	24.0	1.0	6.0	2.0	2.0	1.0	2411.0
293165	40.0	2.0	7.0	2.0	17.0	1.0	2349.0

10644 rows × 11 columns

In [35]:

```
brfss_sub4 = brfss_sub3[brfss_sub2['_RFSMOK3'] != 9]
brfss_sub4
```

```
/tmp/ipython-input-2469558662.py:1: UserWarning: Boolean Series key will be rei
ndexed to match DataFrame index.
  brfss_sub4 = brfss_sub3[brfss_sub2['_RFSMOK3'] != 9]
```

Out[35]:

	_AGE80	SEXVAR	INCOME3	GENHLTH	MENTHLTH	_RFSMOK3	_BMI5
254073	75.0	2.0	4.0	2.0	30.0	1.0	2622.0
254074	18.0	2.0	9.0	2.0	1.0	1.0	2080.0
254077	56.0	1.0	5.0	4.0	3.0	2.0	3261.0
254082	72.0	1.0	7.0	4.0	30.0	1.0	NaN
254083	53.0	1.0	8.0	4.0	10.0	1.0	3839.0
...	...	...	...	...	...	...	...
293160	64.0	2.0	3.0	4.0	2.0	1.0	3411.0
293161	27.0	1.0	4.0	2.0	10.0	1.0	4161.0
293163	33.0	2.0	4.0	3.0	15.0	2.0	4291.0
293164	24.0	1.0	6.0	2.0	2.0	1.0	2411.0
293165	40.0	2.0	7.0	2.0	17.0	1.0	2349.0

9985 rows × 11 columns

In [36]:
```
brfss_sub5 = brfss_sub4[
    (brfss_sub4['CVDINFR4'] != 7) &
    (brfss_sub4['CVDINFR4'] != 9) &
    (brfss_sub4['CVDINFR4'] != 'BLANK')
]
brfss_sub5
```

Out[36]:

	_AGE80	SEXVAR	INCOME3	GENHLTH	MENTHLTH	_RFSMOK3	_BMI5
254073	75.0	2.0	4.0	2.0	30.0	1.0	2622.0
254074	18.0	2.0	9.0	2.0	1.0	1.0	2080.0
254077	56.0	1.0	5.0	4.0	3.0	2.0	3261.0
254082	72.0	1.0	7.0	4.0	30.0	1.0	NaN
254083	53.0	1.0	8.0	4.0	10.0	1.0	3839.0
...	...	...	...	...	...	...	...
293160	64.0	2.0	3.0	4.0	2.0	1.0	3411.0
293161	27.0	1.0	4.0	2.0	10.0	1.0	4161.0
293163	33.0	2.0	4.0	3.0	15.0	2.0	4291.0
293164	24.0	1.0	6.0	2.0	2.0	1.0	2411.0
293165	40.0	2.0	7.0	2.0	17.0	1.0	2349.0

9919 rows × 11 columns

In [37]:
```
brfss_sub6 = brfss_sub5[
    (brfss_sub5['DIABETE4'] != 7) &
    (brfss_sub5['DIABETE4'] != 9) &
    (brfss_sub5['DIABETE4'] != 'BLANK')
]
brfss_sub6
```

Out[37]:

	_AGE80	SEXVAR	INCOME3	GENHLTH	MENTHLTH	_RFSMOK3	_BMI5
254073	75.0	2.0	4.0	2.0	30.0	1.0	2622.0
254074	18.0	2.0	9.0	2.0	1.0	1.0	2080.0
254077	56.0	1.0	5.0	4.0	3.0	2.0	3261.0
254082	72.0	1.0	7.0	4.0	30.0	1.0	NaN
254083	53.0	1.0	8.0	4.0	10.0	1.0	3839.0
...	...	...	...	...	...	...	...
293160	64.0	2.0	3.0	4.0	2.0	1.0	3411.0
293161	27.0	1.0	4.0	2.0	10.0	1.0	4161.0
293163	33.0	2.0	4.0	3.0	15.0	2.0	4291.0
293164	24.0	1.0	6.0	2.0	2.0	1.0	2411.0
293165	40.0	2.0	7.0	2.0	17.0	1.0	2349.0

9911 rows × 11 columns

In [38]:

```
brfss_sub7 = brfss_sub6[
    (brfss_sub6['ADDEPEV3'] != 7) &
    (brfss_sub6['ADDEPEV3'] != 9) &
    (brfss_sub6['ADDEPEV3'] != 'BLANK')
]
brfss_sub7
```

Out[38]:

	_AGE80	SEXVAR	INCOME3	GENHLTH	MENTHLTH	_RFSMOK3	_BMI5
254073	75.0	2.0	4.0	2.0	30.0	1.0	2622.0
254074	18.0	2.0	9.0	2.0	1.0	1.0	2080.0
254077	56.0	1.0	5.0	4.0	3.0	2.0	3261.0
254082	72.0	1.0	7.0	4.0	30.0	1.0	NaN
254083	53.0	1.0	8.0	4.0	10.0	1.0	3839.0
...	...	...	...	...	...	...	...
293160	64.0	2.0	3.0	4.0	2.0	1.0	3411.0
293161	27.0	1.0	4.0	2.0	10.0	1.0	4161.0
293163	33.0	2.0	4.0	3.0	15.0	2.0	4291.0
293164	24.0	1.0	6.0	2.0	2.0	1.0	2411.0
293165	40.0	2.0	7.0	2.0	17.0	1.0	2349.0

9833 rows × 11 columns

In [39]:
```
brfss_sub8 = brfss_sub7[
    (brfss_sub7['TOLDHI3'] != 7) &
    (brfss_sub7['TOLDHI3'] != 9) &
    (brfss_sub7['TOLDHI3'] != 'BLANK')
]
brfss_sub8
```

Out[39]:

	_AGE80	SEXVAR	INCOME3	GENHLTH	MENTHLTH	_RFSMOK3	_BMI5
254073	75.0	2.0	4.0	2.0	30.0	1.0	2622.0
254074	18.0	2.0	9.0	2.0	1.0	1.0	2080.0
254077	56.0	1.0	5.0	4.0	3.0	2.0	3261.0
254082	72.0	1.0	7.0	4.0	30.0	1.0	NaN
254083	53.0	1.0	8.0	4.0	10.0	1.0	3839.0
...	...	...	...	...	...	...	...
293160	64.0	2.0	3.0	4.0	2.0	1.0	3411.0
293161	27.0	1.0	4.0	2.0	10.0	1.0	4161.0
293163	33.0	2.0	4.0	3.0	15.0	2.0	4291.0
293164	24.0	1.0	6.0	2.0	2.0	1.0	2411.0
293165	40.0	2.0	7.0	2.0	17.0	1.0	2349.0

9780 rows × 11 columns

In [40]:
```
brfss_sub_final = brfss_sub8.reset_index(drop=True)
brfss_sub_final
```

Out[40]:

	_AGE80	SEXVAR	INCOME3	GENHLTH	MENTHLTH	_RFSMOK3	_BMI5	CV
0	75.0	2.0	4.0	2.0	30.0	1.0	2622.0	
1	18.0	2.0	9.0	2.0	1.0	1.0	2080.0	
2	56.0	1.0	5.0	4.0	3.0	2.0	3261.0	
3	72.0	1.0	7.0	4.0	30.0	1.0	NaN	
4	53.0	1.0	8.0	4.0	10.0	1.0	3839.0	
...	...	...	...	...	...	...	...	
9775	64.0	2.0	3.0	4.0	2.0	1.0	3411.0	
9776	27.0	1.0	4.0	2.0	10.0	1.0	4161.0	
9777	33.0	2.0	4.0	3.0	15.0	2.0	4291.0	
9778	24.0	1.0	6.0	2.0	2.0	1.0	2411.0	
9779	40.0	2.0	7.0	2.0	17.0	1.0	2349.0	

9780 rows × 11 columns

In [41]:
```
brfss_sub_final = brfss_sub_final.dropna(subset=['_BMI5', 'TOLDHI3'])
```

In [42]:
```
random_5k_rows = brfss_sub_final.sample(n=5000, random_state=42)
random_5k_rows = random_5k_rows.reset_index(drop=True)
random_5k_rows.drop(columns=['DIABETE4'] ,inplace=True)
random_5k_rows
```

Out[42]:

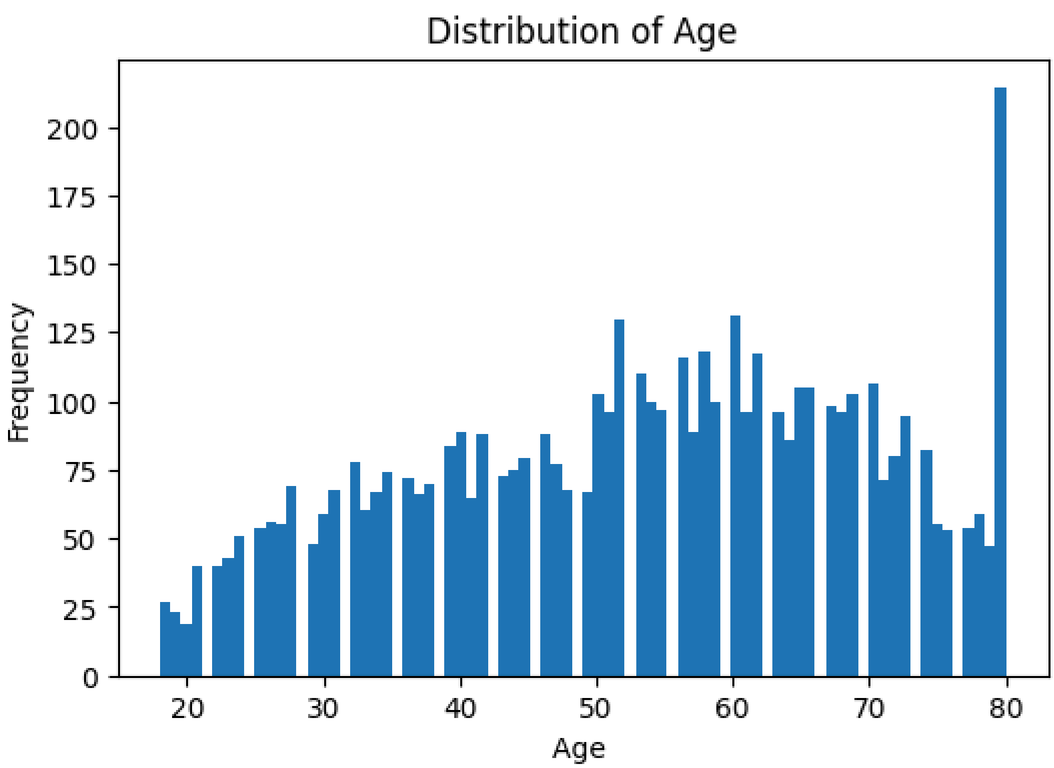
	_AGE80	SEXVAR	INCOME3	GENHLTH	MENTHLTH	_RFSMOK3	_BMI5	CV
0	26.0	1.0	8.0	2.0	2.0	1.0	2584.0	
1	46.0	1.0	11.0	3.0	5.0	1.0	3499.0	
2	74.0	1.0	10.0	2.0	5.0	1.0	2481.0	
3	55.0	1.0	10.0	1.0	3.0	1.0	2580.0	
4	51.0	1.0	9.0	3.0	2.0	1.0	2953.0	
...	...	...	...	...	...	...	...	
4995	80.0	1.0	6.0	4.0	30.0	1.0	2744.0	
4996	25.0	1.0	8.0	2.0	20.0	1.0	2518.0	
4997	72.0	1.0	5.0	4.0	30.0	1.0	1994.0	
4998	73.0	2.0	7.0	3.0	5.0	1.0	2163.0	
4999	56.0	2.0	9.0	2.0	2.0	1.0	3433.0	

5000 rows × 10 columns

In [43]:
```
import pandas as pd


age_80_plus_count = random_5k_rows[random_5k_rows['_AGE80'] == 23].shape[0]
total_count = random_5k_rows.shape[0]
percentage = (age_80_plus_count / total_count) * 100

print(f"Total rows in random_5k_rows: {total_count}")
print(f"Respondents aged 80 or older: {age_80_plus_count}")
print(f"Percentage aged 80 or older: {percentage:.2f}%")
```

```
Total rows in random_5k_rows: 5000
Respondents aged 80 or older: 43
Percentage aged 80 or older: 0.86%
```

In [44]:
```
import matplotlib.pyplot as plt
age_col = random_5k_rows['_AGE80']


plt.figure(figsize=(6,4))
plt.hist(age_col, bins=80)
plt.title("Distribution of Age")
plt.xlabel("Age ")
plt.ylabel("Frequency")
plt.show()

age_col.describe()
```



Out[44]:

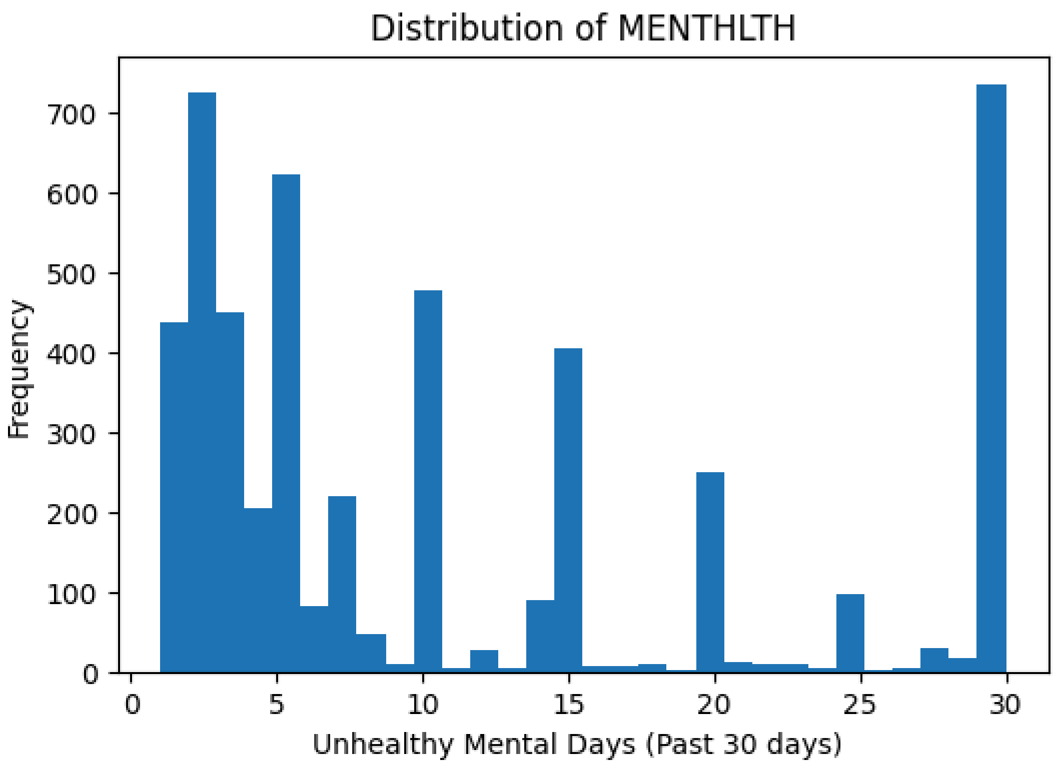
	_AGE80
count	5000.000000
mean	52.860800
std	16.472392
min	18.000000
25%	40.000000
50%	54.000000
75%	66.000000
max	80.000000

dtype: float64

We can see that the most common age is 80 and this is anomaly more than the others, and most of the sample's ages are between 50 and 70 years old.

In [45]:

```
ment_col = random_5k_rows['MENTHLTH']


plt.figure(figsize=(6,4))
plt.hist(ment_col, bins=30)
plt.title("Distribution of MENTHLTH")
plt.xlabel("Unhealthy Mental Days (Past 30 days) ")
plt.ylabel("Frequency")
plt.show()

ment_col.describe()
```



Out[45]:

	MENTHLTH
count	5000.000000
mean	10.862200
std	10.028944
min	1.000000
25%	3.000000
50%	6.000000
75%	15.000000
max	30.000000

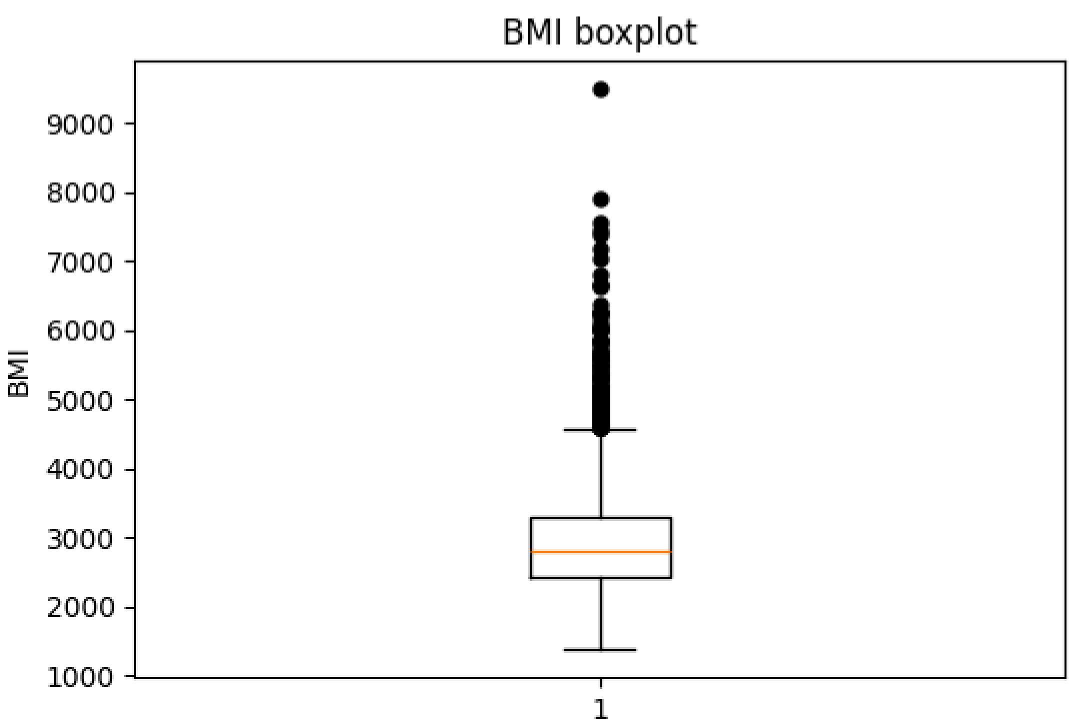
dtype: float64

The number of people reporting poor mental health generally decreases as the number of days increases, but the distribution shows a major spike in reports at 30 days. This pattern is best explained by the all-or-nothing thinking bias: when asked to quantify their experiences over 30 days, most respondents tend to simplify their answer by choosing the extreme categories reporting: either 0 days (everything is good) or 30 days (everything is very bad).

In [46]:

```
bmi_col = random_5k_rows['_BMI5']
```

```
bmi_col = pd.to_numeric(bmi_col, errors='coerce')

bmi_col = bmi_col.dropna()

plt.figure(figsize=(6,4))
plt.boxplot(
    bmi_col,
    showfliers=True,
    flierprops=dict(
        marker='o',
        markersize=5,
        markerfacecolor='black',
        markeredgecolor='black'
    )
)
plt.title("BMI boxplot")
plt.ylabel("BMI")
plt.show()

bmi_col.describe()
```



Out[46]:

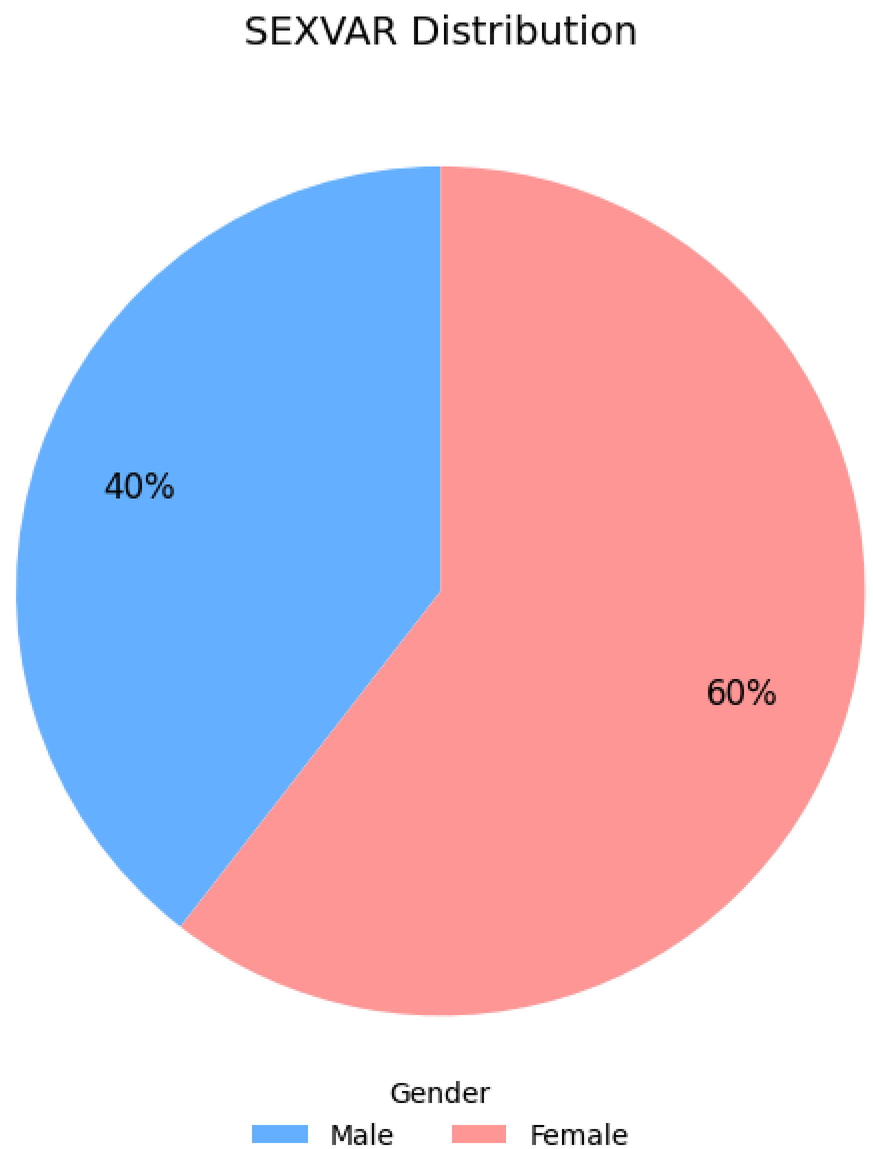
	_BMI5
count	5000.000000
mean	2946.457400
std	748.067828
min	1372.000000
25%	2433.000000
50%	2812.500000
75%	3292.000000
max	9494.000000

dtype: float64

Here we can see that there are a lot of outliers in the high side, it means that specificly in New York there are a lot obese/overweight people.

In [47]:
```
import matplotlib.pyplot as plt
import pandas as pd

sex_col = pd.to_numeric(random_5k_rows['SEXVAR'], errors='coerce').dropna()

mapping = {1: "Male", 2: "Female"}
sex_named = sex_col.map(mapping)

order = ["Male", "Female"]
counts = sex_named.value_counts().reindex(order).fillna(0)

colors = ["#66b3ff", "#ff9999"]
plt.figure(figsize=(6,6))
wedges, texts, autotexts = plt.pie(
    counts,
    autopct="%1.0f%%",
    startangle=90,
    colors=colors,
    textprops={'color': "black", 'fontsize': 12},
    pctdistance=0.75
)

plt.legend(
    wedges,
    counts.index,
    title="Gender",
    loc="lower center",
    bbox_to_anchor=(0.5, -0.05),
    ncol=2,
    frameon=False
)
```

```
plt.title("SEXVAR Distribution", fontsize=14)
plt.tight_layout()
plt.show()
```

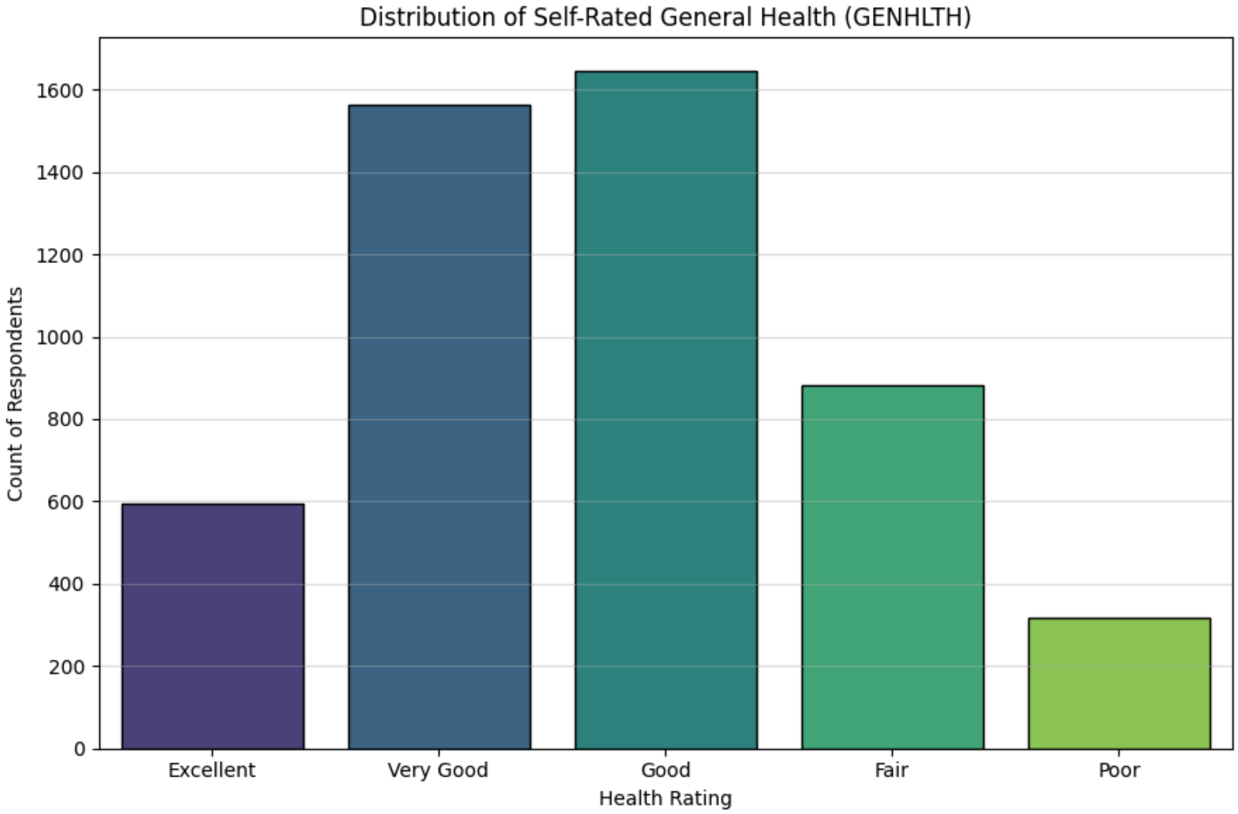


There are 20% more women than men in the sample, and this difference might affect some of our results. We will consider this when we do the rest of the analysis.*italicized text*

In [48]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

health_labels = {
    1.0: "Excellent",
    2.0: "Very Good",
    3.0: "Good",
```

```
    4.0: "Fair",
    5.0: "Poor"
}


random_5k_rows_copy = random_5k_rows.copy()
random_5k_rows_copy['HEALTH_RATING'] = random_5k_rows_copy['GENHLTH'].map(heal

order = list(health_labels.values())

plt.figure(figsize=(9, 6))

sns.countplot(
    x='HEALTH_RATING',
    data=random_5k_rows_copy,
    order=order,
    palette='viridis',
    edgecolor='black'
)

plt.title('Distribution of Self-Rated General Health (GENHLTH)')
plt.xlabel('Health Rating')
plt.ylabel('Count of Respondents')
plt.xticks(rotation=0)
plt.grid(axis='y', alpha=0.4)
plt.tight_layout()

plt.savefig('genhlth_barplot.png')
print("Bar plot saved as genhlth_barplot.png")
```

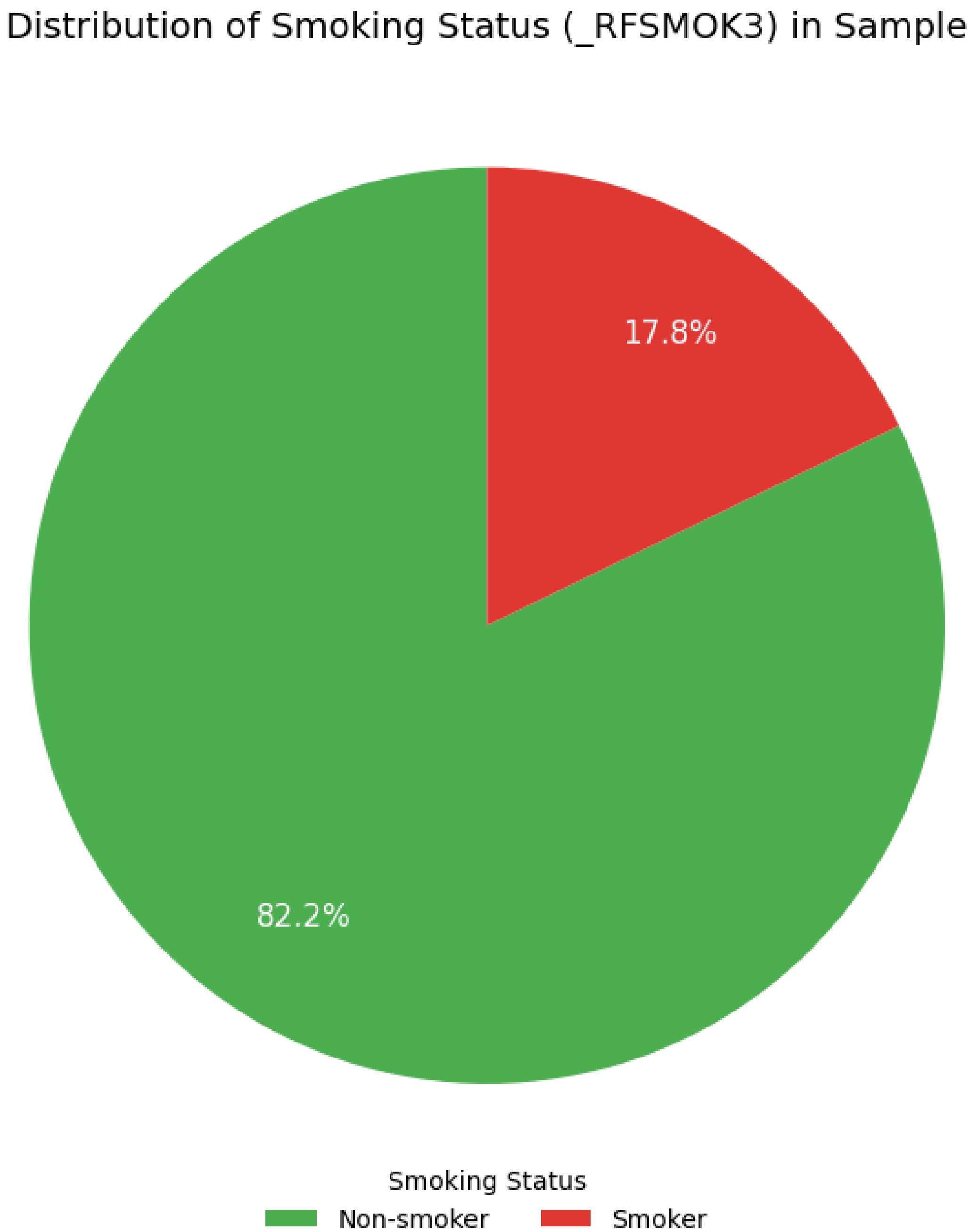
```
/tmp/ipython-input-2173423624.py:21: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same e
ffect.

  sns.countplot(
Bar plot saved as genhlth_barplot.png
```



we can see that the graph skewed to the left, which means most people rated their health as good the bars are highest at "Good" and "Very Good." The data has a long tail going towards the "Poor" rating, this just shows that most of the people in the sample think their health is pretty good.

In [49]:
```
import matplotlib.pyplot as plt
import pandas as pd



smoker_col = pd.to_numeric(random_5k_rows['_RFSMOK3'], errors='coerce').dropna

mapping = {1.0: "Non-smoker", 2.0: "Smoker"}
smoker_named = smoker_col.map(mapping)

order = ["Non-smoker", "Smoker"]
counts = smoker_named.value_counts().reindex(order).fillna(0)

colors = ["#4CAF50", "#E53935"]

plt.figure(figsize=(7,7))
wedges, texts, autotexts = plt.pie(
    counts,
    autopct="%1.1f%%",
    startangle=90,
    colors=colors,
    textprops={'color': "white", 'fontsize': 12},
```

```
    pctdistance=0.75
)

plt.legend(
    wedges,
    counts.index,
    title="Smoking Status",
    loc="lower center",
    bbox_to_anchor=(0.5, -0.05),
    ncol=2,
    frameon=False
)

plt.title("Distribution of Smoking Status (_RFSMOK3) in Sample", fontsize=14)
plt.tight_layout()
plt.show()
```

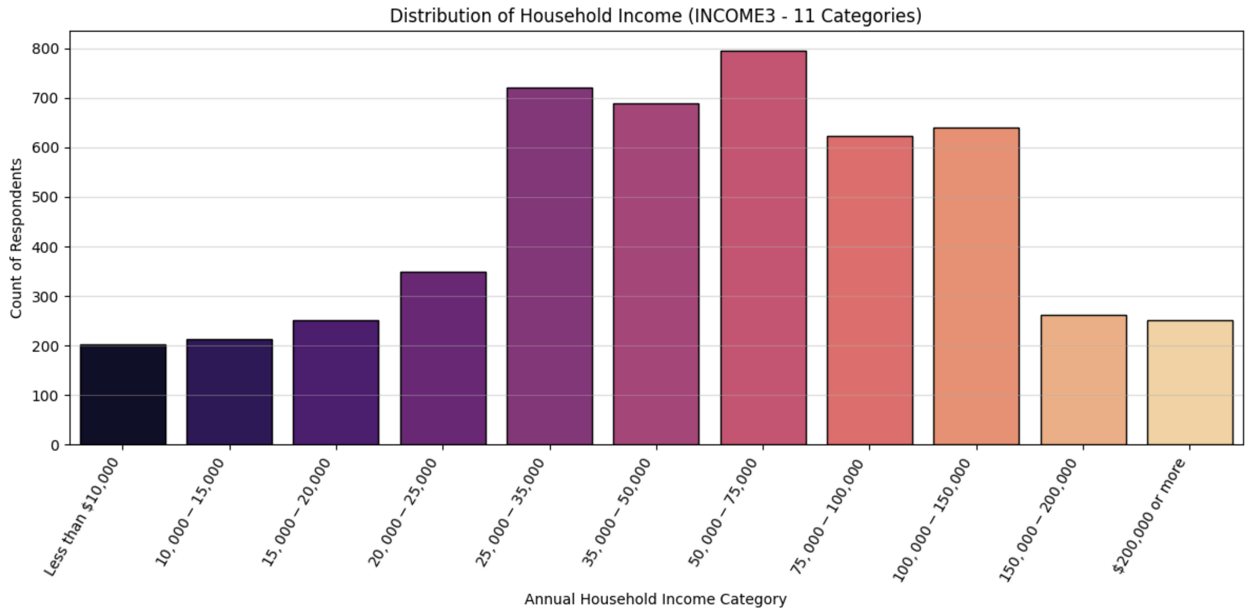


There are close to 4 times more Non-smokers than smokers in the sample, and this difference might affect some of our results. We will consider this when we do the rest of the analysis.

In [50]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

income_labels = {
    1.0: "Less than $10,000",
    2.0: "$10,000 -  $15,000",
    3.0: "$15,000  -  $20,000",
```

```
    4.0: "$20,000  -  $25,000",
    5.0: "$25,000  -  $35,000",
    6.0: "$35,000  -  $50,000",
    7.0: "$50,000  -  $75,000",
    8.0: "$75,000  -  $100,000",
    9.0: "$100,000  - $150,000",
    10.0: "$150,000  - $200,000",
    11.0: "$200,000 or more"
}


random_5k_rows_copy = random_5k_rows.copy()

random_5k_rows_copy['INCOME_RATING'] = random_5k_rows_copy['INCOME3'].map(inco

order = list(income_labels.values())

plt.figure(figsize=(12, 6))
sns.countplot(
    x='INCOME_RATING',
    data=random_5k_rows_copy,
    order=order,
    palette='magma',
    edgecolor='black'
)

plt.title('Distribution of Household Income (INCOME3 - 11 Categories)')
plt.xlabel('Annual Household Income Category')
plt.ylabel('Count of Respondents')
plt.xticks(rotation=60, ha='right')
plt.grid(axis='y', alpha=0.4)
plt.tight_layout()

plt.savefig('income_barplot.png')
print("Bar plot saved as income_barplot.png")
```

```
/tmp/ipython-input-683111991.py:27: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same e
ffect.

  sns.countplot(
Bar plot saved as income_barplot.png
```



In [ ]:

The income graph shows a distribution that is not normal and is slightly skewed to the right. Most respondents are clustered in the middle-to-upper income brackets, with the largest group earning between 50,000 - 75,000 usd, and the second largest between 25,000 - 35,000 usd.

Research questions

A. Does a high BMI lead to increase in the number of days people feel poor mental health?

This is a regression question whose continuous explanatory variable is _BMI5 and whose continuous explained variable is MENTHLTH.

B. Does an increase in a person's age lead to an increase in the probability of having high cholesterol?

This is a regression question whose continuous explanatory variable is _AGE80 and whose binary explained variable is TOLDHI3.

C. Is there a statistically significant difference in the BMI between individuals who have ever been told they had a heart attack and those who have not?

This is the test question where the continuous variable is _BMI5 and the binary variable is CVDINFR4.

More research questions:

1. Does a person's annual income category affect their chance of having depression?
2. Does increasing age affect the chance of having depression?